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中文论文题目：基于平均流模型的一步去噪

英文论文题目：One-step Denoising with MeanFlow Model

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摘要

本工作探索了平均流一步生成式模型 (MeanFlow) 在图像去噪任务中的应用。本研究的主要目标是理解流匹配生成式模型 (Flow Matching) 及其优化方案 MeanFlow 模型的基本原理,并在 MNIST 与 CIFAR-10 数据集上进行实验。在此基础上,我们将 MeanFlow 模型改造用于图像去噪任务:不再以简单的高斯分布作为初始分布,而是以加噪声的真实图像作为起点,通过 MeanFlow 模型的一步生成过程对图像进行去噪。MNIST 实验表明,一次函数评估 ($\text{NFE} = 1$) 即可显著降低噪声,使峰值信噪比 (PSNR) 从约 6 dB 提升至 22.6 dB。二次函数评估 ($\text{NFE} = 2$) 将 PSNR 进一步提升至约 24 dB,而更多推理步骤增益有限。实验结果表明,MeanFlow 模型能以极低的推理成本有效完成图像去噪任务。本工作未提出新的算法或模型框架,而是对一步生成模型在图像去噪任务中的应用进行了实践探索,初步验证了 MeanFlow 方法在快速图像去噪中的可行性。

Abstract

This work investigates the application of the MeanFlow one-step generative model to image denoising tasks. The primary objective of this summer research is to understand the fundamental principles of Flow Matching generative modeling and its improved version, the MeanFlow model, through theoretical derivation and experiments. Based on this understanding, we specifically adapt the MeanFlow model for image denoising by modifying its initial distribution: from simple distributions like Gaussian to noisy images, retaining the model’s one-step generation capacity while serving the denoising purpose. Experiments on MNIST dataset show that a single function evaluation ($\text{NFE} = 1$) significantly reduces the noise, increasing the Peak Signal-to-Noise Ratio (PSNR) from approximately 6 dB to 22.6 dB. A second evaluation ($\text{NFE} = 2$) further improves the PSNR to 24 dB, while additional sampling steps yield limited gains. This work demonstrates that the MeanFlow framework is capable of performing effective image denoising at minimal sampling cost, positioning it as a practical alternative to conventional multi-step denoising methods. Although our work does not propose new algorithms or model frameworks, it explores the application of the MeanFlow framework to image denoising, verifying its feasibility for future image restoration tasks.

1 Introduction

Generative modeling aims to transform a simple prior distribution into a complex target data distribution. Among existing approaches, flow-based models provide an understandable method by explicitly defining an invertible transformation between random variables and computing the corresponding Jacobian determinants. In contrast, the Flow Matching framework [1] achieves this by modeling an instantaneous velocity field that transports one distribution into another based on the continuity equation. Both approaches are built on concepts of “flows” and construct an explicit mapping from the prior to the target distribution. However, they differ in what the transform is and how it is learned.

Recently, research has paid more attention to few-step, and particularly, one-step generative models which offer significant sampling advantages. The MeanFlow model [2] extends Flow Matching by introducing a ground-truth average velocity field derived from the instantaneous velocity field in Flow Matching. The average velocity is defined as the ratio of displacement to a time interval, with the displacement computed as the integral of the instantaneous velocity. This formulation allows the neural network to directly learn from the well-defined average velocity without any extra consistency heuristic [2], enabling reliable and effective one-step generation. Moreover, classifier-free guidance (CFG) can be incorporated into the target vector field without additional cost at sampling time, preserving model’s efficiency.

In this work, we focus on Flow Matching and MeanFlow. Our primary goal is to understand their fundamental principles through theoretical derivation and hands-on experiments. Building on this foundation, we adapt the MeanFlow framework for one-step image denoising task. Specifically, we alter its initial distribution to noisy images instead of gaussian noise, retaining its one-step generation ability and serving the denoising purpose. Experiments on MNIST demonstrate that MeanFlow denoising method can substantially reduce noise with just a single function evaluation, highlighting its potential as an efficient one-step denoising method.

2 Background

2.1 Flow Based Model

Flow-based models define an invertible transformation between a simple prior distribution and a target data distribution.

Let x denote a random variable following a simple prior distribution, such as the standard Gaussian, and let y denote a variable following the target distribution. Then there exists a mapping $f : \mathbb{R}^d \rightarrow \mathbb{R}^d$ such that $y = f(x)$. By the change of variable formula, the probability density function of x and y relates with:

$$p_Y(y) = p_X(x) \frac{\partial x}{\partial y}, \quad (1)$$

where $\frac{\partial x}{\partial y}$ is the Jacobian matrix of the inverse transformation f^{-1} .

With this formula, we can learn the mapping function f directly, and this explicit relationship makes the flow-based model interpretable. However, computing the Jacobian determinant is usually computationally expensive, and this constrains the model design.

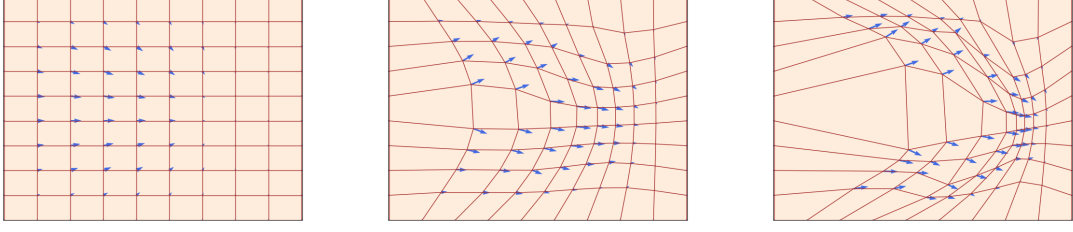


Figure 1: Illustration of flows. Here, the evolution of the density is defined by a velocity field $v : \mathbb{R}^d \rightarrow \mathbb{R}^d$ (blue arrows) specifying its instantaneous movements at all locations (here $d = 2$). Figure from [3].

2.2 Flow Matching

2.2.1 Continuity Equation and Velocity Field

The continuity equation describes how a density evolves with a velocity field. Let $v(x, t)$ and $\rho(x, t)$ denote the velocity field and the density of a substance at position x and time t . The continuity equation is given by:

$$\frac{\partial \rho(x, t)}{\partial t} + \nabla_x \cdot (\rho(x, t)v(x, t)) = 0. \quad (2)$$

It states that, for any infinitesimal volume, the change of the density inside the volume equals the net flux across the boundary. This local balance, under some reasonable assumptions like the convergence of the global density, implies the global conservation of the substance. In our case, the “total substance” corresponds to the probability mass, and $\rho(x, t)$ represents the probability density. As a distribution evolves from a prior to the target data distribution, the continuity equation motivates the use of the velocity field to describe this evolution.

Let p_{prior} and p_{target} denote the prior and target distribution, and let $x_0 \sim p_{\text{prior}}$ and $x_1 \sim p_{\text{target}}$. Suppose the distribution evolves from $t = 0$ to $t = 1$ with intermediate distribution p_t and $x_t \sim p_t$, we defined the velocity field $v(x_t, t)$ as:

$$v(x_t, t) \triangleq \frac{dx_t}{dt}. \quad (3)$$

Given $v(x_t, t)$, a sample from p_t can be obtained by:

$$x_t - x_0 = \int_0^t v(x_\tau, \tau) d\tau. \quad (4)$$

In particular, selecting $t = 1$ produces a sample from p_{target} .

2.2.2 Probability Path and the Training Target

A deterministic trajectory from p_{prior} to p_{target} is called a probability path or marginal probability path. This evolution is driven by the marginal velocity field $v(x_t, t)$. However, directly obtaining $v(x_t, t)$ is generally difficult. Flow Matching constructs an equivalent velocity $v^{\text{tgt}}(x_t, t)$ with the weighted average.

Consider sampling z from p_{target} and defining a conditional probability path from p_{prior} to δ_z with conditional distribution $p_t(\cdot | x)$:

$$x_t = (1 - t)x_0 + tz. \quad (5)$$

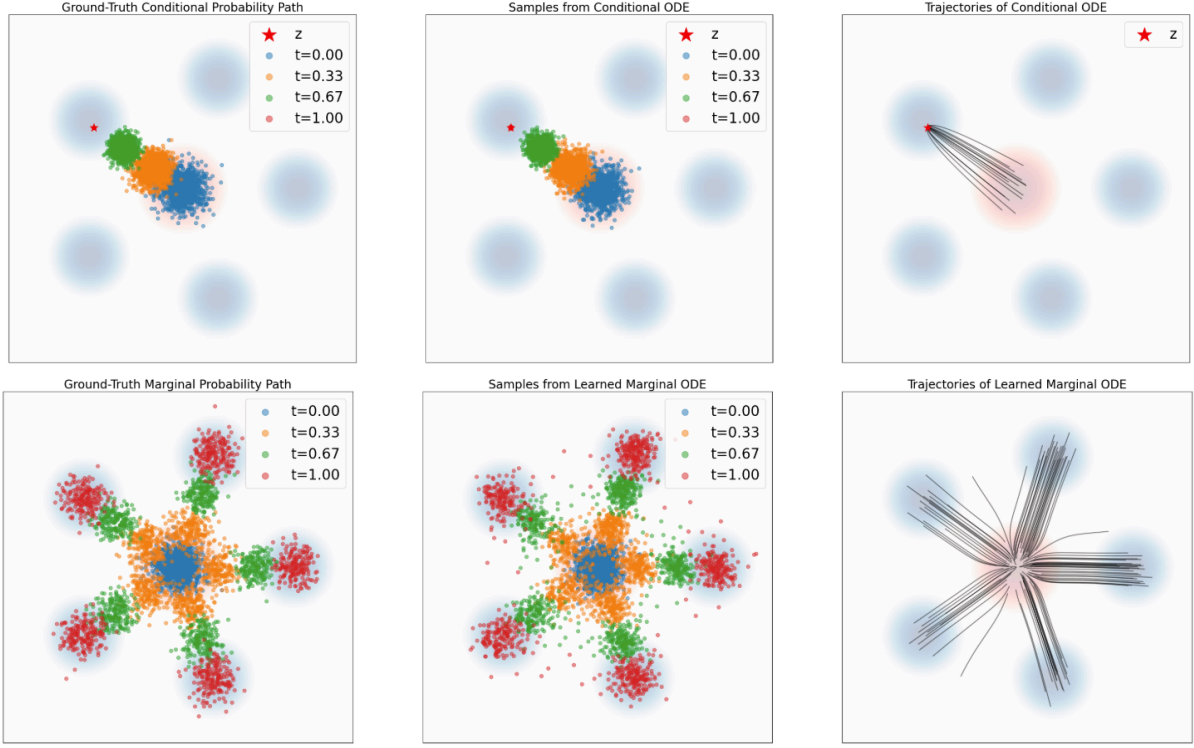


Figure 2: Illustration of conditional and marginal probability path. Target distribution p_{target} in blue background. Gaussian prior distribution p_{prior} in red background. Top row: Conditional probability path. Left: Ground truth samples from $p_t(\cdot | z)$. Middle: ODE samples over times. Right: Trajectories by simulating ODE in Equation (6). Bottom row: Simulating a marginal probability path. Left: Ground truth samples from p_t . Middle: ODE samples over times. Right: Trajectories by simulating ODE in Equation (3). As one can see, the conditional vector field follows the conditional probability path and the marginal vector field follows the marginal probability path. Figure from [3].

The marginal distribution is then given by the total probability thorem:

$$p_t(x_t) = \int p_t(x_t | x) p_{\text{target}}(x) dx.$$

The corresponding conditional velocity field is:

$$v(x_t, t | z) \triangleq \frac{dx_t}{dt} = z - x_0. \quad (6)$$

The marginal velocity, which serves as the training target, is constructed by taking the expectation of the conditional velocity under the weights of z :

$$\begin{aligned} v^{\text{tgt}}(x_t, t) &\triangleq \int v(x_t, t | z) p_t(z | x_t) dx_t \\ &= \int v(x_t, t | z) \frac{p_t(x_t | z) p_{\text{target}}(z)}{p_t(x_t)} dz. \end{aligned} \quad (7)$$

The validity of this construction with respect to $v^{\text{tgt}}(x_t, t)$ is discussed in Appendix A.

2.2.3 Train the Model

Our generative model is trained by fitting a neural network $v^\theta(x_t, t)$ to approximate the target velocity field $v^{\text{tgt}}(x_t, t)$. A natural choice of objective is the mean squared error (MSE):

$$\mathcal{L}_{\text{FM}}(\theta) = \mathbb{E}_{t, x_t \sim p_t} \left[\|v^\theta(x_t, t) - v^{\text{tgt}}(x_t, t)\|_2^2 \right] \quad (8)$$

However, there is no closed-form expression for $v^{\text{tgt}}(x_t, t)$, and what we have access to is only samples from p_{target} and p_{prior} . We now show that, under the MSE loss, one may equivalently replace $v^{\text{tgt}}(x_t, t)$ with the conditional velocity field $v(x_t, t \mid z)$:

$$\mathcal{L}_{\text{CFM}}(\theta) = \mathbb{E}_{t, z \sim p_{\text{target}}, x_t \sim p_t(\cdot \mid z)} \left[\|v^\theta(x_t, t) - v^{\text{tgt}}(x_t, t \mid z)\|_2^2 \right], \quad (9)$$

and the two objectives share the same gradients:

$$\nabla_\theta \mathcal{L}_{\text{CFM}}(\theta) = \nabla_\theta \mathcal{L}_{\text{FM}}(\theta) \quad (10)$$

Proof. Expanding the square in $\mathcal{L}_{\text{FM}}$ and $\mathcal{L}_{\text{CFM}}$, we obtain

$$\begin{aligned} \mathcal{L}_{\text{FM}}(\theta) &= \mathbb{E}_{t, x_t \sim p_t} \left[\|v^\theta(x_t, t)\|_2^2 \right] - 2\mathbb{E}_{t, x_t \sim p_t} \left[\|v^\theta(x_t, t)^\top v^{\text{tgt}}(x_t, t)\|_2 \right] + \mathbb{E}_{t, x_t \sim p_t} \left[\|v^{\text{tgt}}(x_t, t)\|_2^2 \right] \\ \mathcal{L}_{\text{CFM}}(\theta) &= \mathbb{E}_{t, z \sim p_{\text{target}}, x_t \sim p_t(\cdot \mid z)} \left[\|v^\theta(x_t, t)\|_2^2 \right] - 2\mathbb{E}_{t, z \sim p_{\text{target}}, x_t \sim p_t(\cdot \mid z)} \left[v^\theta(x_t, t)^\top v^{\text{tgt}}(x_t, t \mid z) \right] \\ &\quad + \mathbb{E}_{t, z \sim p_{\text{target}}, x_t \sim p_t(\cdot \mid z)} \left[\|v^{\text{tgt}}(x_t, t \mid z)\|_2^2 \right] \end{aligned}$$

For the cross term, we compute

$$\begin{aligned} \mathbb{E}_{t, x_t \sim p_t} \left[v^\theta(x_t, t)^\top v^{\text{tgt}}(x_t, t) \right] &= \iint v^\theta(x_t, t)^\top v^{\text{tgt}}(x_t, t) p_t(x_t) dt dx_t \\ &= \iint v^\theta(x_t, t)^\top \left(\int v(x_t, t \mid z) \frac{p_t(x_t \mid z) p_{\text{target}}(z)}{p_t(x_t)} dz \right) p_t(x_t) dt dx_t \\ &= \iiint v^\theta(x_t, t)^\top v(x_t, t \mid z) p_t(x_t \mid z) p_{\text{target}}(z) dz dt dx_t \\ &= \mathbb{E}_{t, z \sim p_{\text{target}}, x_t \sim p_t(\cdot \mid z)} \left[v^\theta(x_t, t)^\top v(x_t, t \mid z) \right] \end{aligned}$$

The expectation of $v^\theta(x_t, t)$ are identical under two different measures, thus $\mathcal{L}_{\text{FM}}$ and $\mathcal{L}_{\text{CFM}}$ differ only in terms independent of θ . Since such terms vanish under gradient, $\nabla_\theta \mathcal{L}_{\text{CFM}}(\theta) = \nabla_\theta \mathcal{L}_{\text{FM}}(\theta)$.

More generally, for any error function that is linear in $v^{\text{tgt}}(x_t, t)$, one may substitute $v^{\text{tgt}}(x_t, t)$ with $v(x_t, t \mid z)$ without affecting the optimization (see Appendix B).

3 Methodology

3.1 Mean Flow

Following [2], the instantaneous velocity field in Flow Matching can be replaced with an average velocity to enable one-step generation. Specifically, from time r to time t , the average velocity is defined as:

$$u(x_t, r, t) \triangleq \frac{1}{t-r} \int_r^t v(x_\tau, \tau) d\tau. \quad (11)$$

Rewrite Equation (11) as $(t-r)u(x_t, r, t) = \int_r^t v(x_\tau, \tau) d\tau$, and differentiate both sides with respect to t :

$$u(x_t, r, t) = v(x_t, t) - (t-r) \frac{d}{dt} u(x_t, r, t),$$

Applying the total derivative theorem, we obtain:

$$\begin{aligned} u(x_t, r, t) &= v(x_t, t) - (t-r) \left(\frac{\partial u}{\partial x_t} \frac{dx_t}{dt} + \frac{\partial u}{\partial r} \frac{dr}{dt} + \frac{\partial u}{\partial t} \frac{dt}{dt} \right) \\ &= v(x_t, t) - (t-r) \left(\frac{\partial u}{\partial x_t} v(x_t, t) + \frac{\partial u}{\partial t} \right) \end{aligned} \quad (12)$$

since $dr/dt = 0$ and $dx_t/dt = v(x_t, t)$.

Similar to Flow Matching, we train a neural network $u^\theta(x_t, r, t)$ to approximate $u(x_t, r, t)$ with an MSE loss:

$$\mathcal{L}_{\text{MF}}(\theta) = \mathbb{E}_{t,r,x_t \sim p_t} \left[\|u^\theta(x_t, r, t) - \text{sg}[u^{\text{tgt}}(x_t, r, t)]\|_2^2 \right], \quad (13)$$

where $\text{sg}[\cdot]$ denotes the stop gradient operator and $u^{\text{tgt}}(x_t, r, t)$ is given by:

$$u^{\text{tgt}}(x_t, r, t) = v^{\text{tgt}}(x_t, t) - (t-r) \left(\frac{\partial u^\theta}{\partial x_t} v^{\text{tgt}}(x_t, t) + \frac{\partial u^\theta}{\partial t} \right). \quad (14)$$

Here, the derivatives of u are approximated by those of u^θ , following [2].

Since the error term $u^\theta(x_t, r, t) - \text{sg}[u^{\text{tgt}}(x_t, r, t)]$ is a linear function of $v^{\text{tgt}}(x_t, t)$, and according to the result in Appendix B, this allows $v^{\text{tgt}}(x_t, t)$ to be substituted with the conditional velocity $v(x_t, t | z)$:

$$\mathcal{L}_{\text{CMF}}(\theta) = \mathbb{E}_{t,r,z \sim p_{\text{target}}, x_t \sim p_t(\cdot|z)} \left[\|u^\theta(x_t, r, t) - \text{sg}[u_{\text{cond}}^{\text{tgt}}(x_t, r, t)]\|_2^2 \right] \quad (15)$$

$$u_{\text{cond}}^{\text{tgt}} = v(x_t, t | z) - (t-r) \left(\frac{\partial u^\theta}{\partial x_t} v(x_t, t | z) + \frac{\partial u^\theta}{\partial t} \right) \quad (16)$$

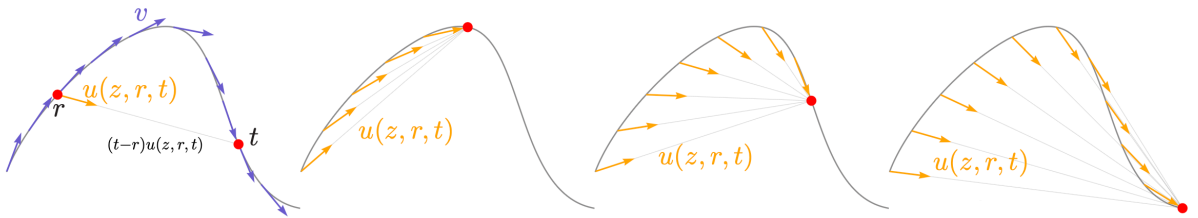


Figure 3: Illustration of the average velocity $u(z, r, t)$. Here, z means the current location, i.e. x_t in Equation (11). Leftmost: The instantaneous velocity $v(x_t, t)$. Right three subplots: The average velocity field at different time t . Figure revised from [2].

This formulation allows MeanFlow to be trained similar to Flow Matching, while directly targeting an average velocity field that facilitates the one-step generation.

Conclusion

This is the conclusion.

References

- [1] Yaron Lipman, Ricky T. Q. Chen, Heli Ben-Hamu, Maximilian Nickel, and Matt Le. Flow matching for generative modeling, 2023.
- [2] Zhengyang Geng, Mingyang Deng, Xingjian Bai, J. Zico Kolter, and Kaiming He. Mean flows for one-step generative modeling, 2025.
- [3] Peter Holderrieth and Ezra Erives. An introduction to flow matching and diffusion models, 2025.

Appendix

A Verification of the Constructed Marginal Velocity

Under our construction $x_t = (1-t)x_0 + tz$, the trajectory of each sample is deterministic and given by a simple affine transformation. Such deterministic transform preserves the total probability mass: the density is only shifted and rescaled. Consequently, the evolution of $p_t(x_t | z)$ must satisfy the continuity equation:

$$\frac{\partial}{\partial t} p_t(x_t | z) + \nabla_{x_t} \cdot (p_t(x_t | z) \cdot v(x_t, t | z)) = 0.$$

Consider the marginal probability density:

$$\begin{aligned} \frac{\partial}{\partial t} p_t(x_t) &= \frac{\partial}{\partial t} \int p_t(x_t | z) p_{\text{target}}(z) dz \\ &= \int \frac{\partial}{\partial t} p_t(x_t | z) p_{\text{target}}(z) dz. \end{aligned}$$

Substitute $\frac{\partial}{\partial t} p_t(x_t | z)$ with the continuity equation:

$$\begin{aligned} \frac{\partial}{\partial t} p_t(x_t) &= \int -\nabla_{x_t} \cdot (p_t(x_t | z) \cdot v(x_t, t | z)) p_{\text{target}}(z) dz \\ &= - \int \nabla_{x_t} \cdot (p_t(x_t | z) \cdot v(x_t, t | z) p_{\text{target}}(z)) dz \\ &= -\nabla_{x_t} \cdot \int v(x_t, t | z) \cdot p_t(x_t | z) p_{\text{target}} dz \\ &= -\nabla_{x_t} \cdot \int v(x_t, t | z) \frac{p_t(x_t | z) p_{\text{target}}(z)}{p_t(x_t)} p_t(x_t) dz \\ &= -\nabla_{x_t} \cdot \left(\left(\int v(x_t, t | z) \frac{p_t(x_t | z) p_{\text{target}}(z)}{p_t(x_t)} dz \right) \cdot p_t(x_t) \right) \\ &= -\nabla_{x_t} \cdot (v^{\text{tgt}}(x_t, t) \cdot p_t(x_t)). \end{aligned}$$

B Conclusion of the Linear-Form Error

We show that whenever the error term is a linear function of the target velocity $v^{\text{tgt}}(x_t, t)$, the same training gradient can be obtained by substituting it with the conditional velocity $v(x_t, t | z)$. Formally, assume

$$\text{error} = Av^{\text{tgt}}(x_t, t) - b_\theta,$$

where A is a fixed matrix independent of the learnable parameters θ , and b_θ is a function of θ .

Define the marginal and conditional objectives as:

$$\mathcal{L}_{\text{marginal}} = \mathbb{E}_{t, x_t \sim p_t} \left[\|b_\theta - Av^{\text{tgt}}(x_t, t)\|_2^2 \right] \quad (17)$$

$$\mathcal{L}_{\text{conditional}} = \mathbb{E}_{t, z \sim p_{\text{target}}, x_t \sim p_t(\cdot | z)} \left[\|b_\theta - Av(x_t, t | z)\|_2^2 \right] \quad (18)$$

As in Section 2.2.3, the expectations of $\|b_\theta\|_2^2$ under both measures coincide. Besides, $\|Av^{\text{tgt}}(x_t, t)\|_2^2$ and $\|Av^{\text{tgt}}(x_t, t)\|_2^2$

are independent of θ . And we show that:

$$\begin{aligned}
\mathbb{E}_{t, x_t \sim p_t} [b_\theta^\top A v^{\text{tgt}}(x_t, t)] &= \iint b_\theta^\top A v^{\text{tgt}}(x_t, t) p_t(x_t) dt dx_t \\
&= \iint b_\theta^\top A \left(\int v(x_t, t | z) \frac{p_t(x_t | z) p_{\text{target}}(z)}{p_t(x_t)} dz \right) p_t(x_t) dt dx_t \\
&= \iiint b_\theta^\top A v(x_t, t | z) p_t(x_t | z) p_{\text{target}}(z) dz dt dx_t \\
&= \mathbb{E}_{t, z \sim p_{\text{target}}, x_t \sim p_t(\cdot | z)} [b_\theta^\top A v(x_t, t | z)]
\end{aligned}$$

This shows that $\mathcal{L}_{\text{marginal}}$ and $\mathcal{L}_{\text{conditional}}$ yield identical gradients with respect to θ whenever the error depends linearly on $v^{\text{tgt}}(x_t, t)$.